

EV BANK

SECURITY LAB / CTF INVESTIGATION REPORT

APK Reconnaissance → Web Investigation → Credential Recovery → Legacy Vault

Prepared by
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Investigation Duration
1 hour 30 minutes

Report Date
09 August 2026

Controlled security-lab exercise. All findings documented in this report were obtained from the supplied EV BANK APK and the associated lab web application.

Executive Summary

This report documents a structured five-stage investigation of the EV BANK training application. The assessment followed the intended lab chain: static APK reconnaissance, discovery of a hidden administration route, recovery of developer credentials from the APK, inspection of the internal administration dashboard, and recovery of a Base64-encoded legacy vault artifact.

Stage	Result	Status
Task 1 — APK Recon	Public server: https://ev-bank-web.edgeone.dev/	COMPLETED
Task 2 — Hidden Admin	Route: /admin.html	COMPLETED
Task 3 — Credentials	Username: admin Password: passadmin123	COMPLETED
Task 4 — Admin Trail	Legacy reference: EVB-LEGACY-07	COMPLETED
Task 5 — Final Vault	assets/legacy_vault.json → Base64 → final flag	COMPLETED

Investigation Chain

APK	Public URL	Admin Route	Credentials	Legacy Ref.	Artifact	Decode	Flag
↓	↓	↓	↓	↓	↓	↓	✓

Final Flag

EV{EV_BANK_LEGACY_VAULT_COMPLETE}

Methodology & Tooling

The investigation was performed as a controlled lab exercise. The APK was decoded with Apktool and inspected in Visual Studio Code. The associated web application was reviewed through the browser, including page-source inspection. Static-analysis findings were then used to authenticate to the lab administration portal. The final artifact was located by searching for the legacy reference and decoded according to the encoding identified in the artifact.

Tool / Interface	Purpose
Apktool	Decode APK resources, manifest, smali and assets for static inspection.
Visual Studio Code	Search decoded APK files for URLs, credentials, references and artifacts.
Web Browser	Inspect public site, view HTML source and access the lab administration portal.
PowerShell / Base64 decoding	Decode the Base64 payload contained in the legacy vault artifact.

90-Minute Activity Timeline

Time	Activity	Output
0–20 min	APK reconnaissance and URL discovery	Public EV BANK server URL
20–35 min	Website source inspection	Hidden /admin.html route
35–55 min	APK credential recovery	Developer admin username/password
55–70 min	Admin dashboard investigation	EVb-LEGACY-07
70–90 min	Legacy artifact search and decoding	Final flag

TASK 1

APK Reconnaissance — Public Web Infrastructure

Objective

Identify the public web infrastructure / server URL used by the EV BANK application through static APK analysis.

Server URL	https://ev-bank-web.edgeone.dev/
APK location	MainActivity.smali / DeveloperConfig.smali
Technique	Decoded APK searched for HTTP/HTTPS references in Visual Studio Code.

Evidence 1 — Decompiled APK showing the development server URL

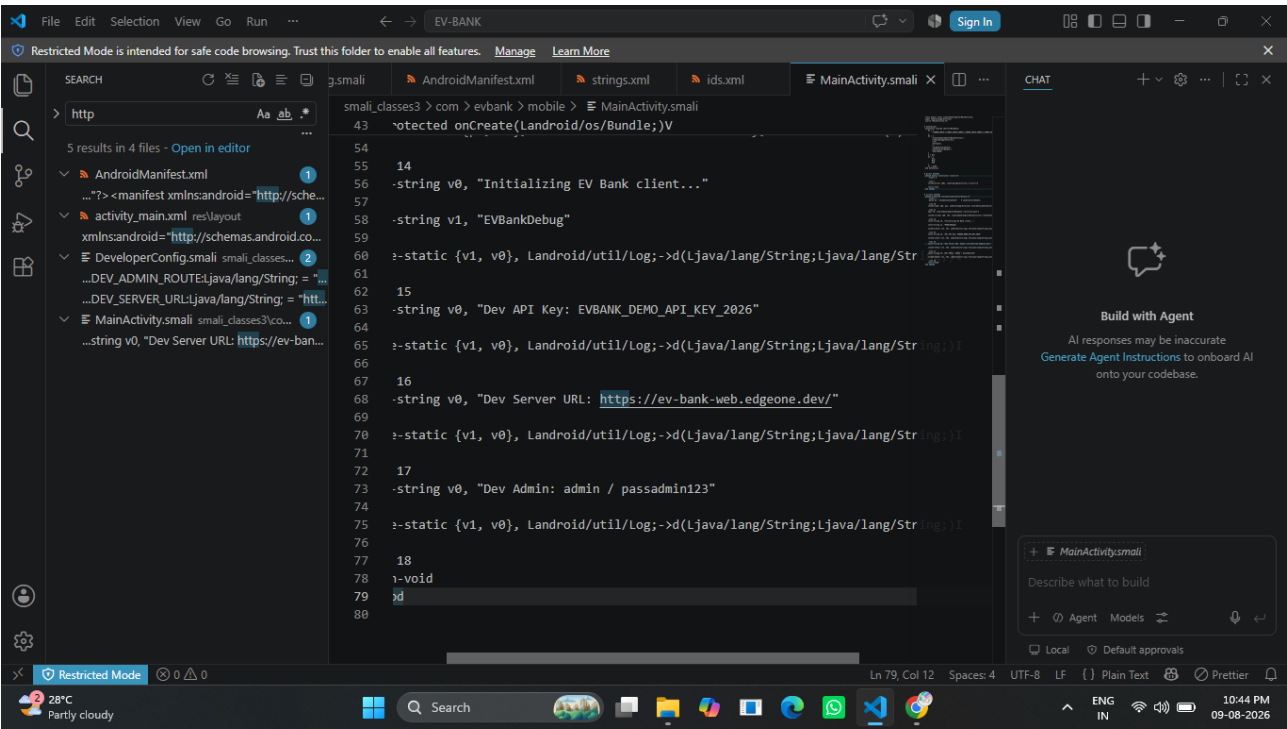


Figure 1. VS Code search/decompiled smali evidence. The visible line identifies the EV BANK development server URL.

Conclusion: Task 1 established the public EV BANK web infrastructure as <https://ev-bank-web.edgeone.dev/>.

TASK 2

Hidden Administration Panel Discovery

Objective

Investigate the public website and identify the hidden internal administration route from the page source.

Admin route	/admin.html
Full route	https://ev-bank-web.edgeone.dev/admin.html
Discovery method	Browser → View Page Source → search for 'admin'.

Evidence 2 — Homepage HTML source

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>EV BANK - Digital Banking Infrastructure</title>
  <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&display=swap" rel="stylesheet">
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <!-- EVB internal console reference: /admin.html -->

  <header>
    <div class="logo"> EV BANK</div>
    <nav>
      <ul>
        <li><a href="#home">Home</a></li>
        <li><a href="#services">Services</a></li>
        <li><a href="#security">Security</a></li>
        <li><a href="#about">About</a></li>
      </ul>
    </nav>
  </header>

  <main id="home">
    <!-- Hero Section -->
    <section class="hero">
      <h1>EV BANK</h1>
      <p>Digital Banking Infrastructure</p>
      <p style="font-size: 1.1rem; color: #94a3b8; max-width: 600px; margin: -1.5rem auto 2.5rem auto;">
        Secure infrastructure for modern digital banking.
      </p>
      <a href="#services" class="btn">Explore Platform</a>
    </section>
  </main>
</body>
</html>
```

Figure 2. HTML source contains the internal console reference: /admin.html.

Conclusion: The source-code comment exposed the internal administration portal route without requiring route guessing.

TASK 3

Developer Credential Recovery

Objective

Recover the administrator credentials through static analysis of the supplied APK.

Credential	Recovered value	Source
Username	admin	DeveloperConfig.smali
Password	passadmin123	DeveloperConfig.smali

Evidence 3A — Developer configuration

```
DEV_ADMIN_PASSWORD:Ljava/lang/String; = "passadmin123"
DEV_ADMIN_ROUTE:Ljava/lang/String; = "https://ev-bank-web.edgeone.dev/admin.h
DEV_ADMIN_USERNAME:Ljava/lang/String; = "admin"
DEV_API_KEY:Ljava/lang/String; = "EVBANK_DEMO_API_KEY_2026"
DEV_SERVER_URL:Ljava/lang/String; = "https://ev-bank-web.edgeone.dev/"
INSTANCE:Lcom/evbank/mobile/DeveloperConfig;

clinit>()V
```

Figure 3. Decompiled DeveloperConfig.smali showing the hardcoded developer password, admin route, username and server URL.

Evidence 3B — Credential lines clearly visible

```
final DEV_ADMIN_PASSWORD:Ljava/lang/String; = "passadmin123"
final DEV_ADMIN_ROUTE:Ljava/lang/String; = "https://ev-bank-web.edgeone.dev/
final DEV_ADMIN_USERNAME:Ljava/lang/String; = "admin"
final DEV_API_KEY:Ljava/lang/String; = "EVBANK_DEMO_API_KEY_2026"
final DEV_SERVER_URL:Ljava/lang/String; = "https://ev-bank-web.edgeone.dev/"
final INSTANCE:Lcom/evbank/mobile/DeveloperConfig;
```

Figure 4. Decompiled configuration showing DEV_ADMIN_USERNAME = admin and DEV_ADMIN_PASSWORD = passadmin123.

Conclusion: The administrator credential was recovered directly from the APK configuration rather than guessed or brute-forced.

TASK 4

Admin Dashboard Investigation

Objective

Use the recovered credentials to access the internal dashboard and identify the developer's legacy reference.

Evidence 4A — Successful admin dashboard access

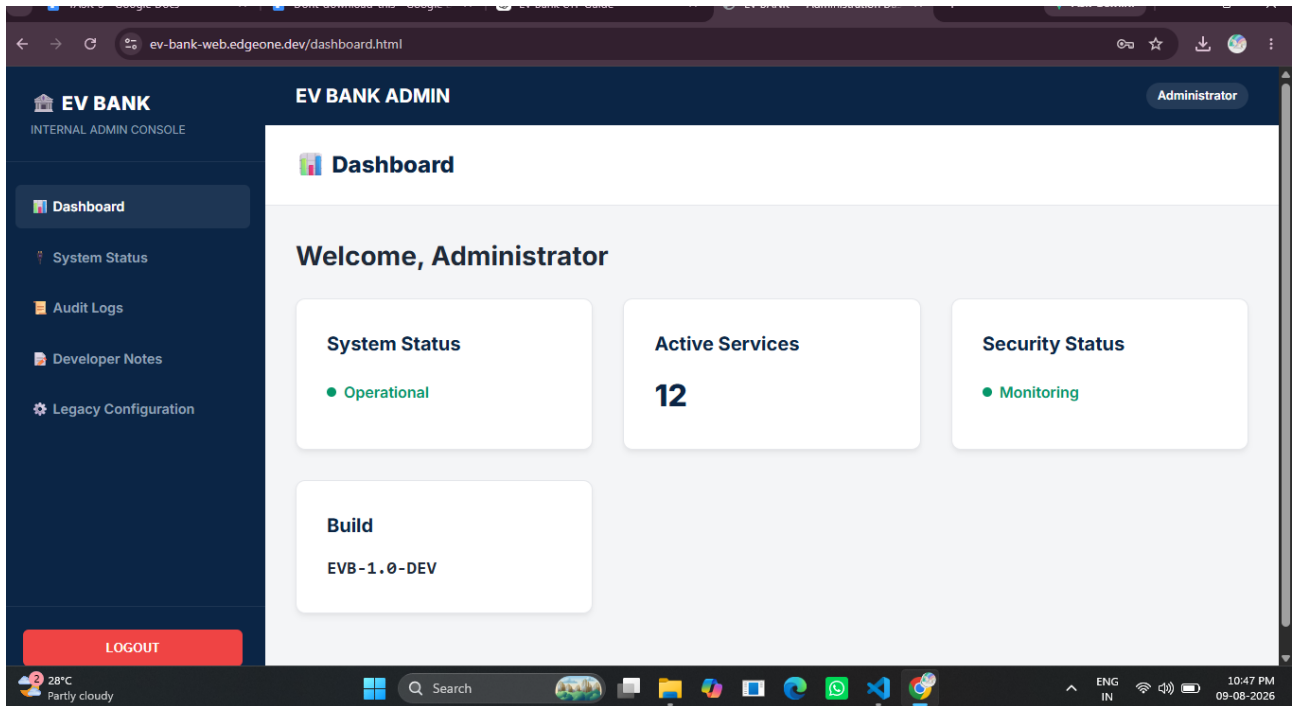


Figure 5. EV BANK Admin dashboard after successful authentication.

Evidence 4B — Developer Notes and legacy reference

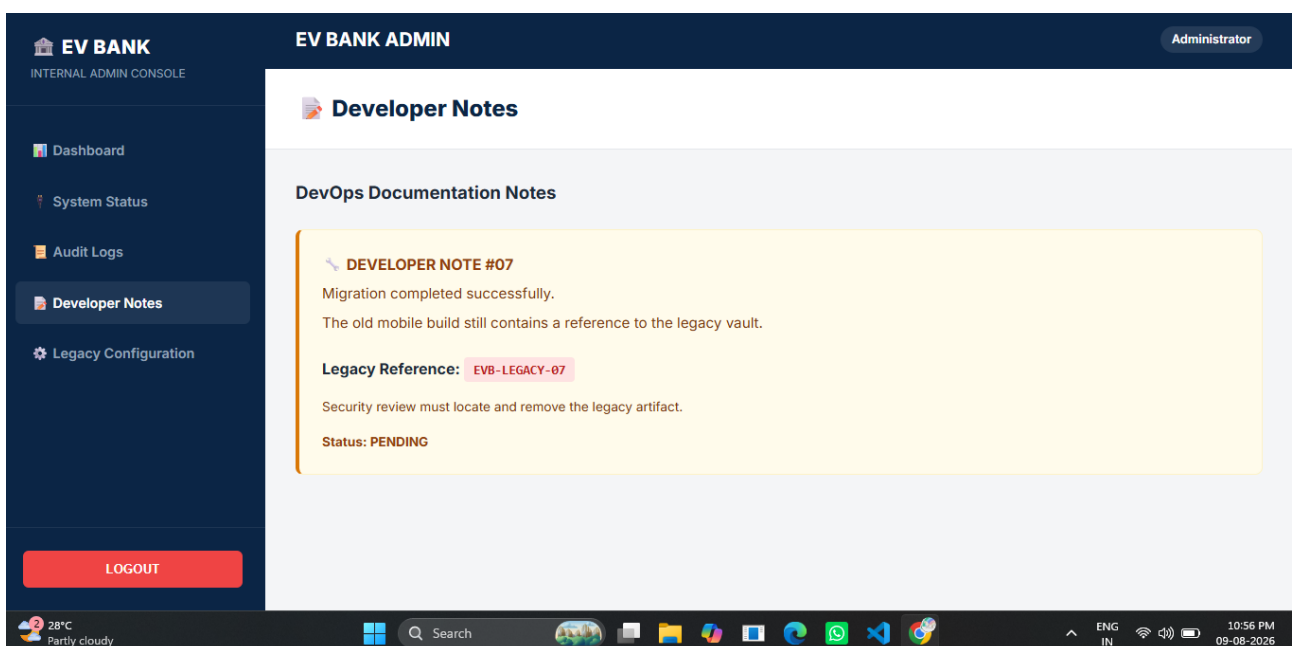


Figure 6. Developer Note #07 states that the old mobile build still contains a reference to the legacy vault and identifies EVB-LEGACY-07.

Finding	Result
Dashboard section	Developer Notes
Developer note	Migration completed successfully; the old mobile build still contains a reference to the legacy vault.
Legacy reference	EVB - LEGACY - 07
Status	PENDING

Conclusion: EVB-LEGACY-07 provided the bridge from the web administration dashboard back to the legacy artifact stored in the APK.

TASK 5

Final Vault — Legacy Artifact & Decoding

Objective

Locate the legacy artifact using EVB-LEGACY-07, identify its encoding, decode the payload and recover the final flag.

Evidence 5A — Legacy artifact discovery

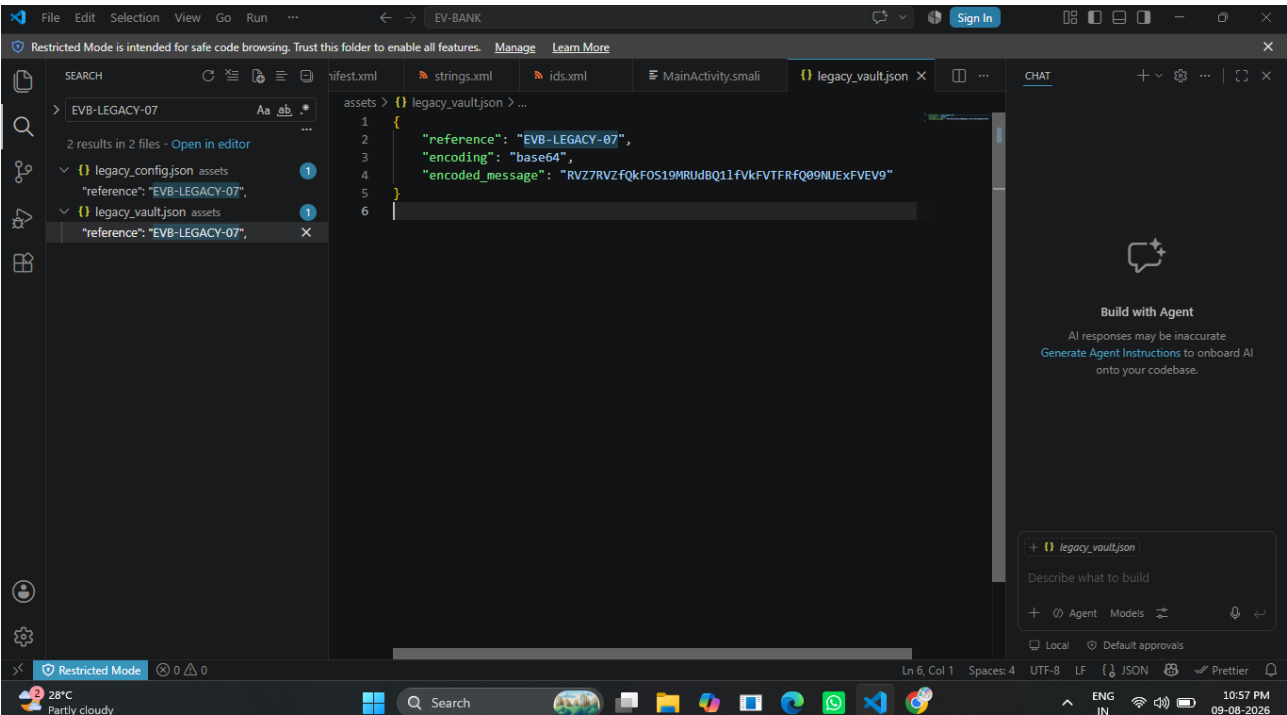


Figure 7. VS Code search for EVB-LEGACY-07 identifies assets/legacy_vault.json. The artifact declares Base64 encoding and contains an encoded_message value.

Item	Finding
Artifact path	assets/legacy_vault.json
Reference	EVB - LEGACY - 07
Encoding	Base64
Encoded message	RVZ7RVZfQkFOS19MRUdBQ11fVkfVTFRfQ09NUExFVEV9
Decoded message	EV{EV_BANK_LEGACY_VAULT_COMPLETE}

Final Result

EV{EV_BANK_LEGACY_VAULT_COMPLETE}

The encoded payload supplied by the legacy vault artifact was Base64-decoded to the final EV BANK lab flag shown above. The artifact itself and the decoded result complete the intended Task 5 chain.

Final Findings & Submission Summary

Task	Required submission item	Final finding
1	Server URL / location / tool / evidence	https://ev-bank-web.edgeone.dev/ — MainActivity.smali / DeveloperConfig.smali — Apktool + VS Code
2	Admin route / discovery / evidence	/admin.html — homepage HTML source
3	Username / password / APK location / evidence	admin / passadmin123 — DeveloperConfig.smali
4	Dashboard section / note / legacy reference / evidence	Developer Notes / Note #07 / EVB-LEGACY-07
5	Artifact / encoding / decoded message / final flag / evidence	assets/legacy_vault.json / Base64 / EV{EV_BANK_LEGACY_VAULT_COMPLETE}

Overall Conclusion

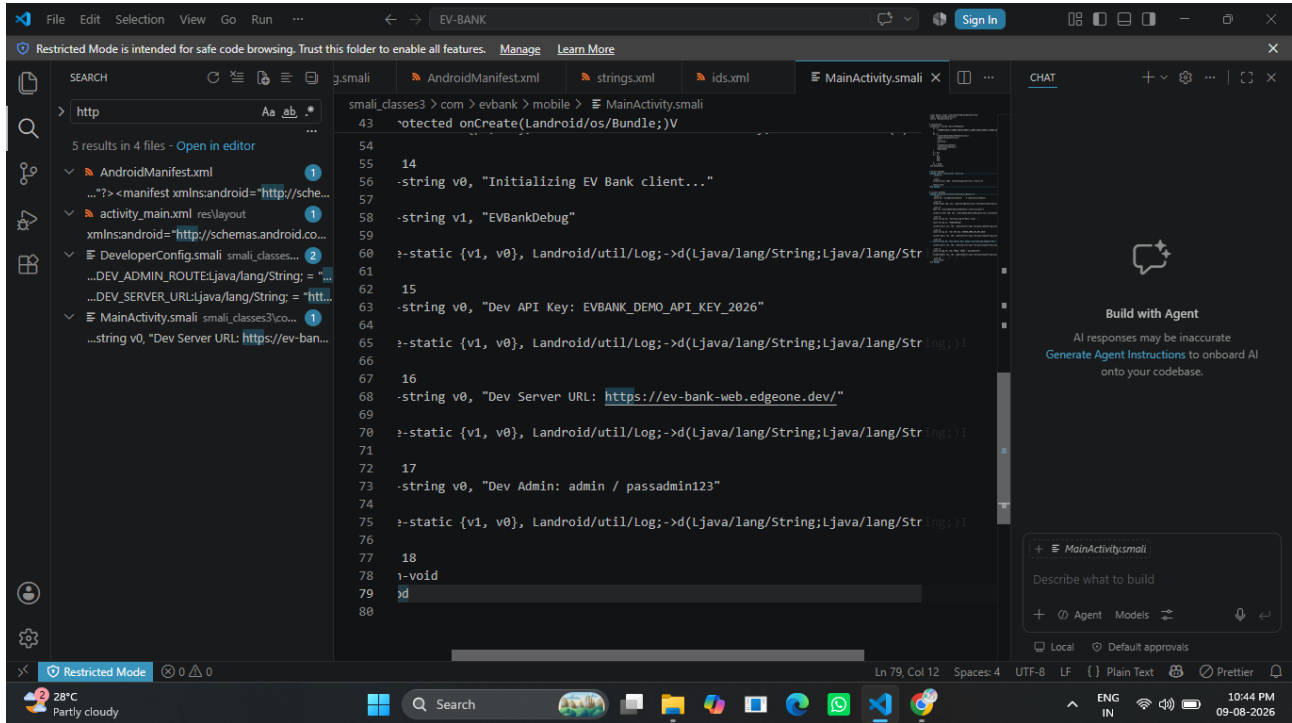
All five EV BANK lab objectives were completed in sequence. The investigation demonstrated how information in a mobile application's static resources can expose public infrastructure, internal routes and developer configuration, and how a web-admin clue can be used to locate a legacy artifact back inside the application package. The final Base64 decoding step recovered the lab flag.

Final Flag	EV{EV_BANK_LEGACY_VAULT_COMPLETE}
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Evidence Appendix — Submitted Screenshots

The following appendix includes the evidence screenshots supplied during the investigation. Each figure is labeled according to the task it supports.

Evidence 1 — Task 1: APK server URL



Screenshot 2026-08-09 224421.png

Evidence 2 — Task 2: Hidden /admin.html route in HTML source

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>EV BANK — Digital Banking Infrastructure</title>
  <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&display=swap" rel="stylesheet">
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <!-- EVB Internal console reference: admin.html -->

  <header>
    <div class="logo"> EV BANK</div>
    <nav>
      <ul>
        <li><a href="#home">Home</a></li>
        <li><a href="#services">Services</a></li>
        <li><a href="#security">Security</a></li>
        <li><a href="#about">About</a></li>
      </ul>
    </nav>
  </header>

  <main id="home">
    <!-- Hero Section -->
    <section class="hero">
      <h1>EV BANK</h1>
      <p>Digital Banking Infrastructure</p>
      <p style="font-size: 1.1rem; color: #94a3b8; max-width: 600px; margin: -1.5rem auto 2.5rem auto;">
        Secure infrastructure for modern digital banking.
      </p>
      <a href="#services" class="btn">Explore Platform</a>
    </section>
  </main>
</body>
</html>
```

Screenshot 2026-08-09 225327.png

Evidence 3A — Task 3: Developer configuration

```
DEV_ADMIN_PASSWORD:Ljava/lang/String; = "passadmin123"  
DEV_ADMIN_ROUTE:Ljava/lang/String; = "https://ev-bank-web.edgeone.dev/admin.h  
DEV_ADMIN_USERNAME:Ljava/lang/String; = "admin"  
DEV_API_KEY:Ljava/lang/String; = "EVBANK_DEMO_API_KEY_2026"  
DEV_SERVER_URL:Ljava/lang/String; = "https://ev-bank-web.edgeone.dev/"  
INSTANCE:Lcom/evbank/mobile/DeveloperConfig;  
  
clinit>()V
```

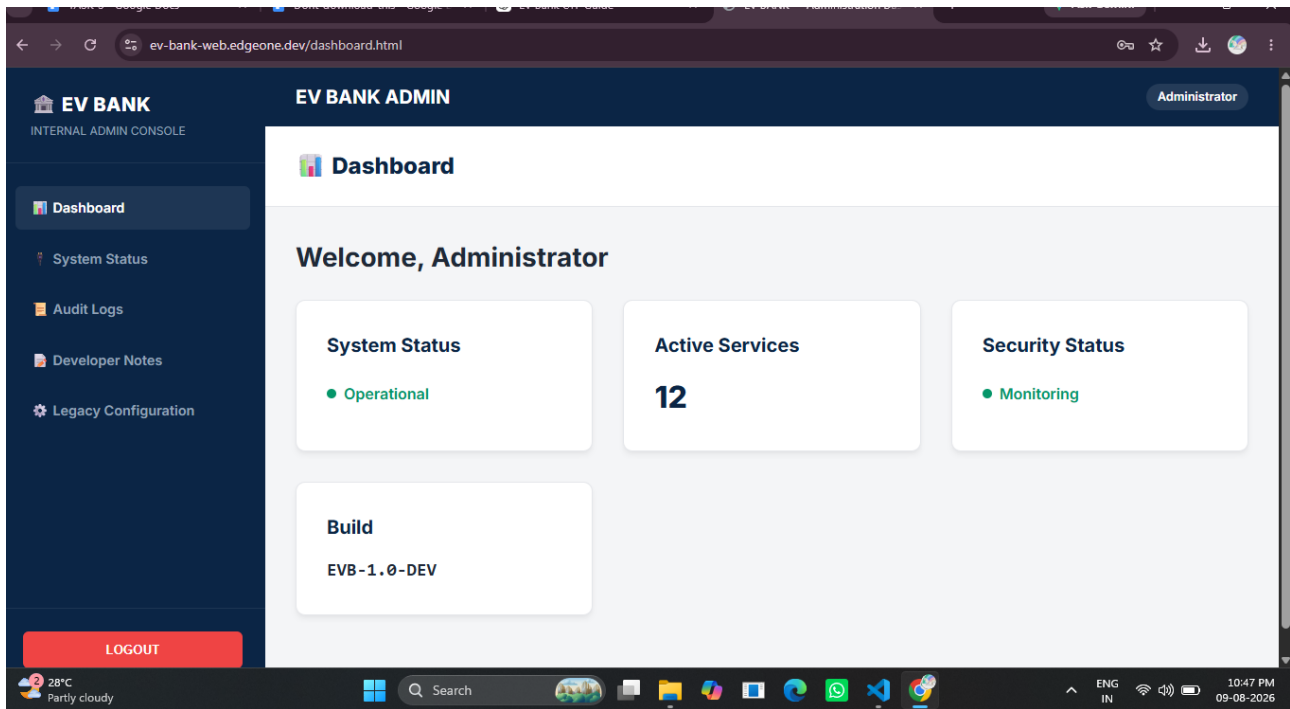
Screenshot 2026-08-09 224509.png

Evidence 3B — Task 3: Username and password lines

```
final DEV_ADMIN_PASSWORD:Ljava/lang/String; = "passadmin123"  
final DEV_ADMIN_ROUTE:Ljava/lang/String; = "https://ev-bank-web.edgeone.dev/  
final DEV_ADMIN_USERNAME:Ljava/lang/String; = "admin"  
final DEV_API_KEY:Ljava/lang/String; = "EVBANK_DEMO_API_KEY_2026"  
final DEV_SERVER_URL:Ljava/lang/String; = "https://ev-bank-web.edgeone.dev/"  
final INSTANCE:Lcom/evbank/mobile/DeveloperConfig;
```

Screenshot 2026-08-09 224633.png

Evidence 4A — Task 4: Admin dashboard



Screenshot 2026-08-09 224726.png

Evidence 4B — Task 4: Developer Notes / EVB-LEGACY-07

The screenshot displays the 'EV BANK ADMIN' interface. The top header bar is dark blue with 'EV BANK ADMIN' on the left and 'Administrator' on the right. Below this, the main content area is titled 'Developer Notes'. On the left side, there is a dark blue sidebar with a menu containing 'Dashboard', 'System Status', 'Audit Logs', 'Developer Notes' (which is highlighted), and 'Legacy Configuration'. At the bottom of the sidebar is a red 'LOGOUT' button. The main content area shows 'DevOps Documentation Notes' with a yellow box containing the following text: 'DEVELOPER NOTE #07', 'Migration completed successfully.', 'The old mobile build still contains a reference to the legacy vault.', 'Legacy Reference: EVB-LEGACY-07', 'Security review must locate and remove the legacy artifact.', and 'Status: PENDING'. The bottom of the screen shows a Windows taskbar with a search bar, various application icons, and system status information including '28°C Partly cloudy', 'ENG IN', and the date '10:56 PM 09-08-2026'.

EV BANK
INTERNAL ADMIN CONSOLE

EV BANK ADMIN Administrator

Developer Notes

DevOps Documentation Notes

DEVELOPER NOTE #07
Migration completed successfully.
The old mobile build still contains a reference to the legacy vault.

Legacy Reference: EVB-LEGACY-07

Security review must locate and remove the legacy artifact.

Status: PENDING

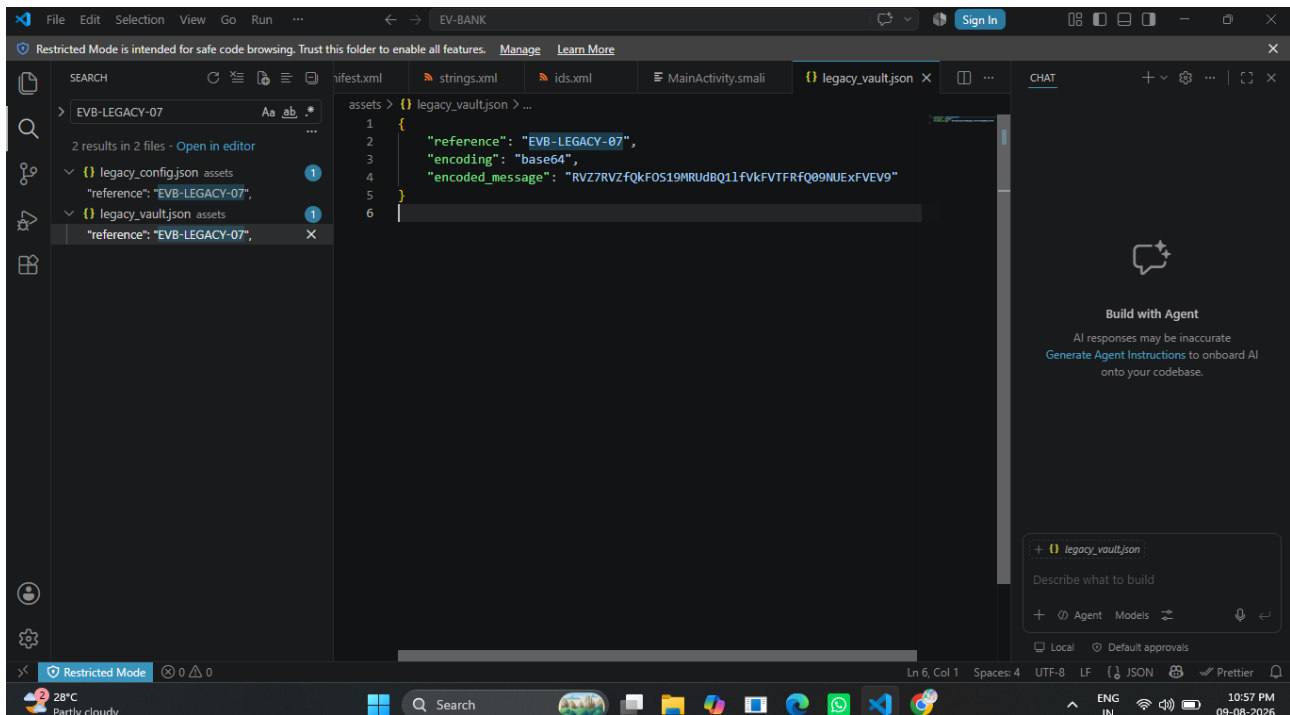
28°C Partly cloudy

Search

ENG IN 10:56 PM 09-08-2026

Screenshot 2026-08-09 225619.png

Evidence 5A — Task 5: legacy_vault.json and Base64 payload



Screenshot 2026-08-09 225752.png

Report Identification

Candidate	DINESH S.
Email	s90960665@gmail.com
Lab	EV BANK Security Lab
Duration	1 hour 30 minutes
Date	09 August 2026
Completion	All 5 tasks completed